DESIGN AND ANALYSIS OF ALGORITHMS

PROJECT REPORT

ShiftNest

Employee Scheduling and Wage Optimization

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Abstract

Employee scheduling requires a company to match staffing needs with employee skills, availability and working-hour limits. A low-cost assignment may be unsuitable when it creates overlapping shifts, insufficient rest or an uneven distribution of work. ShiftNest is an employee scheduling and wage optimization system developed to demonstrate how algorithm design can support this practical decision-making problem. It combines a working web application with scheduling and wage calculations implemented in C17.

The scheduling model uses binary decisions to represent whether an employee is assigned to a shift. Each requirement specifies a time interval, a required skill and a staffing count. The system enforces full-shift availability, skill eligibility, active employment status, non-overlapping assignments, weekly hour limits and a configurable minimum rest period. Managers can create custom requirement names and skills, maintain employee records, generate proposed schedules and make validated assignment changes. Employees have a separate interface for viewing published assignments, submitting availability and acknowledging notifications.

Branch and Bound is the exact optimization method used for small demonstration instances. It explores feasible assignments, computes an optimistic lower bound and prunes branches that cannot improve the best complete solution. A greedy baseline constructs a quick schedule and is followed by a limited local improvement step. When complete, this baseline supplies an initial upper bound for exact search. Runtime and search-node limits preserve responsiveness while retaining the best valid solution found. The application distinguishes proven optimality, unproven feasibility, proven infeasibility and failure to find a complete solution within the limit.

The objective combines normalized estimated wages with workload imbalance. Fairness is measured through the variance of assigned-to-preferred-hour ratios, allowing employees with different preferred workloads to be compared proportionally. Wage calculations use integer minutes and paise, with weekly overtime and documented rounding rules. The same calculation module supports optimization and wage reporting. PostgreSQL stores accounts, availability, requirements, schedule versions and notifications, while server-side authorization separates manager and employee access.

Small deterministic fixtures and exhaustive enumeration checks support evaluation of correctness. The project demonstrates the tradeoff between the speed of greedy scheduling and the stronger guarantees of exact search. Its estimates are based on scheduled hours; attendance-based payroll, statutory deductions and external payment processing remain outside the project scope.

Keywords Employee scheduling, Branch and Bound, greedy algorithm, local search, workload fairness, wage estimation.

Algorithms and Methodology

Scheduling formulation

Each employee-shift decision is either assigned or not assigned. A complete solution fills every required position while satisfying all hard constraints. Among complete solutions, ShiftNest minimizes estimated wages divided by a fixed INR 1000 reference, plus a configurable weight multiplied by workload-ratio variance. This normalization makes the cost-fairness tradeoff explicit; it does not reduce the wage estimate shown to users.

Greedy construction and local improvement

Requirements are ordered by the number of eligible employees relative to staffing demand, placing more constrained shifts first. Greedy construction selects the eligible employee giving the lowest resulting objective at each position. Up to three local improvement passes try valid single-employee replacements and accept only strict objective improvements without reducing coverage. These methods are useful for a quick baseline but do not guarantee a global optimum.

Branch and Bound search

The search state contains current assignments, the next staffing position and the best complete solution found. Depth-first recursion adds a feasible employee, explores subsequent positions and then backtracks. A complete greedy schedule acts as the initial upper bound. The lower bound adds current wages to independent minimum base-wage estimates for remaining positions, with a conservative rounding allowance. Ignoring competition, future overtime and nonnegative fairness penalties keeps this bound optimistic. Infeasible or non-improving branches are pruned; equivalent permutations of identical positions are also avoided.

Complexity and practical limits

With E employees and P staffing positions, the naive search can explore on the order of E to the power P combinations, plus constraint and bound calculations. Pruning often reduces observed work, but exponential worst-case growth remains. Greedy and local improvement require polynomial work for the fixed number of passes. Time and node limits make larger demonstrations practical, although a timeout cannot prove optimality or infeasibility. An incomplete fallback remains clearly marked and cannot be published.

Implementation and verification

The C17 server uses libmicrohttpd for HTTP, cJSON for JSON, libpq for PostgreSQL and libsodium for password hashing and secure tokens. Publication and notification creation occur in one transaction. Verification includes 200 deterministic small instances compared with exhaustive enumeration, tests for skills, availability, rest, hour limits and overtime, and browser checks of publication and employee acknowledgement. The application workflow was verified using local PostgreSQL; the Neon cloud connection remained unverified because the endpoint reset connections.